

```
// Pure Synthia Automata – state-space
constants (wheel math, scales, dimensions,
trigrams, PRNG)

// Wheel precision constants (user's DMS
system, spec §2)
export const WHEEL_ARCSECONDS = 1296000; //
360° in arc-seconds
export const GATE_ARCSECONDS = 20250;    //
5°37'30" per gate (64 gates)
export const LINE_ARCSECONDS = 3375;     //
0°56'15" per line (6 lines/gate)
export const COLOR_ARCSECONDS = 562.5;   //
6 colors/line
export const TONE_ARCSECONDS = 93.75;    //
6 tones/color
export const BASE_ARCSECONDS = 18.75;    //
5 bases/tone

// Finest sub-division: each base splits
into 99 arc-units. 99 does NOT divide
// evenly into a base (18.75/99 =
0.18939...), a gate, or the full 1,296,000"
// wheel – verified, no integer arcsecond
alignment exists at this level.
// arcUnit is therefore a pure positional
index (0-98), not a whole-arcsecond
// count; see
fineIndexForAddress/addressForFineIndex in
addressing.js.
```

```
export const ARC_UNITS_PER_BASE = 99;
export const ZODIAC_ARCSECONDS = 108000; //
30° per sign (12 signs)
export const HOUSE_ARCSECONDS = 162000; //
45° per trigram house (8 houses)

// Scale ladder L0→L8+ (spec §1); index =
level
export const SCALES =
['feature','phoneme','grapheme','morpheme',
'word','phrase','clause','sentence','discou
rse','automaton','mesh'];

// Five dimensional perspectives (spec §3);
D3 Being is the hinge/witness
export const DIMENSIONS =
['Movement','Evolution','Being','Design','S
pace'];
export const DIMENSION_META = {
  Movement: { interrogative:'Where',
operation:'transition', seedGate:1,
octave:2 },
  Evolution:{ interrogative:'What',
operation:'transform', seedGate:2,
octave:3 },
  Being:    { interrogative:'When',
operation:'instantiate', seedGate:6,
octave:4 },
  Design:   { interrogative:'Why',
operation:'structure', seedGate:14,
octave:5 },
  Space:    { interrogative:'Who',
operation:'integrate', seedGate:20,
octave:6 },
```

```

};

// Fu Xi order, bits bottom-to-top (line 1
// first), value = binary (bit0 = bottom line)
export const TRIGRAMS = [
  { id:'kun', name:'Receptive',
    element:'Earth', bits:[0,0,0], value:0
  },
  { id:'zhen', name:'Arousing',
    element:'Thunder', bits:[1,0,0], value:1
  },
  { id:'kan', name:'Abysmal',
    element:'Water', bits:[0,1,0], value:2
  },
  { id:'dui', name:'Joyous',
    element:'Lake', bits:[1,1,0], value:3
  },
  { id:'gen',
    name:'KeepingStill',element:'Mountain',
    bits:[0,0,1], value:4 },
  { id:'li', name:'Clinging',
    element:'Fire', bits:[1,0,1], value:5
  },
  { id:'xun', name:'Gentle',
    element:'Wind', bits:[0,1,1], value:6
  },
  { id:'qian', name:'Creative',
    element:'Heaven', bits:[1,1,1], value:7
  },
];

// Deterministic seeded PRNG factory;
// returns () => [0,1)
export function mulberry32(seed) {

```

```
let a = seed >>> 0;
return function () {
  a |= 0; a = (a + 0x6D2B79F5) | 0;
  let t = Math.imul(a ^ (a >>> 15), 1 |
a);
  t = (t + Math.imul(t ^ (t >>> 7), 61 |
t)) ^ t;
  return ((t ^ (t >>> 14)) >>> 0) /
4294967296;
};
}
```